

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Ship 10+ production features end-to-end across full stack: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io workflows, document processing; 394 commits, 7 deployments. - Own full backend stack (PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry); debug and resolve production issues across all layers.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-based genomic data platform on Google Cloud Run (GCP) with JBrowse 2 genome browser; 100+ researchers, 500GB datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved processing accuracy by 25%. - Automated deployments on AWS and GCP; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Demand Paging	Xinu OS – Systems Programming	2025
- Implemented 8-store virtual memory system for Xinu OS in C: page table management, demand paging, TLB fault handler; reduced page fault rate by 25%. - Engineered 7+ kernel syscalls (address translation, TLB invalidation, frame allocation); improved memory throughput by 15%.		
KubeServe	Kubernetes LLM Inference Platform	2026
- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N pods), Kopf CRD operator for multi-provider serving; circuit breaker + EWMA LB. - Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		
VectorLift	Semantic Search Engine	2026
- Built semantic search in Python on MS MARCO: BM25, dense bi-encoder (in-batch negatives), cross-encoder re-ranker; Kafka-driven Qdrant corpus indexing pipeline. - Scored NDCG@10, MRR@10, MAP; built FastAPI serving with per-stage latency tracking and Streamlit dashboard.		
AlgoPulse	Agentic AI Platform	2025
- Built agentic AI orchestration on Temporal.io and Kafka; sustained 2,000 async inference req/sec across reasoning pipelines. - Replaced API polling with Supabase Edge Functions streaming; eliminated state loss under peak load.		

Languages and Technologies

- Languages:** C, Python, TypeScript, JavaScript, Java, SQL, Bash
- Systems & Performance:** Linux, Kubernetes, KEDA, Docker, OpenTelemetry, Prometheus, Grafana, k6
- Backend & Data:** FastAPI, PostgreSQL, Redis, Kafka, Spark, Hadoop, Temporal.io, REST APIs
- AI/ML:** Claude/GPT-4o/Gemini APIs, LangChain, RAG, LLM evaluation, PyTorch, prompt engineering

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog